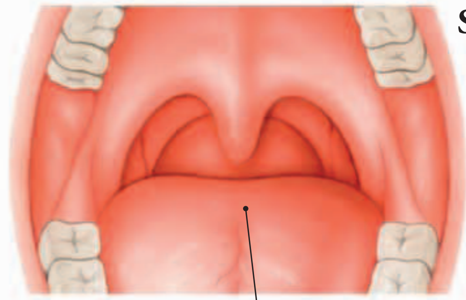


Tongue muscles help shape sounds.

SHAPING SPEECH

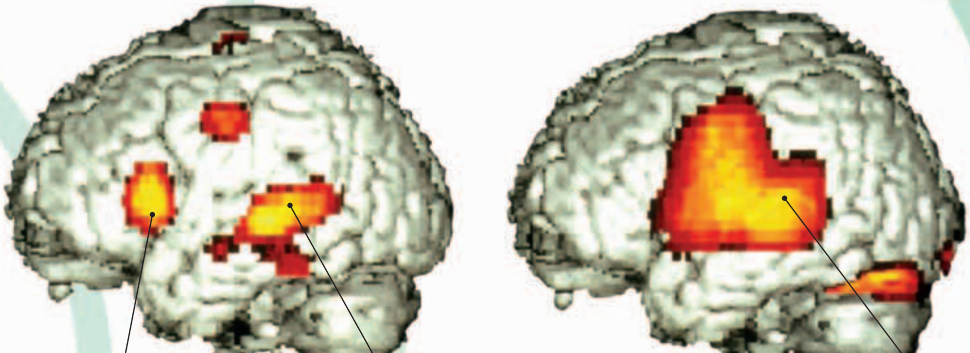
Under orders from your brain, the muscles controlling your tongue, lips, and cheeks shape the sounds coming from your vocal cords. They create a sequence of words, spoken in a recognizable language, which reveals your thoughts and feelings to others.

FAST FACTS

- Around 6,900 languages are spoken in the world today. The top five most spoken languages are Mandarin Chinese, English, Spanish, Hindi, and Russian.
- The vocal cords of a soprano—a woman with a high singing voice—vibrate at up to 2,000 times a second.
- The vocal cords of a bass—a man with a low singing voice—vibrate just 60 times a second.

IN CONTROL

These two scans show which areas of your brain are used when you speak (left) and when you listen to spoken words (right). In the first scan, Wernicke's area finds the correct words to match what the brain has to say and sends signals to Broca's area. This, in turn, sends signals to the voice box and the mouth instructing them to produce speech.



Broca's area controls speech.

Speaking

Wernicke's area "understands" words.

Listening

The hearing area of the brain

MAKING MUSIC

Some people can use their breathing systems to make sounds much louder than their own voices. Controlled bursts of air from this musician's lungs create buzzing vibrations in her lips. These sound vibrations enter and travel through the long tube-shaped trumpet, which makes the sounds much louder.

▶ HEARING *Sound waves from the speaker's mouth enter the listener's ears. Here they are detected by sound receptors that send signals to the brain.*